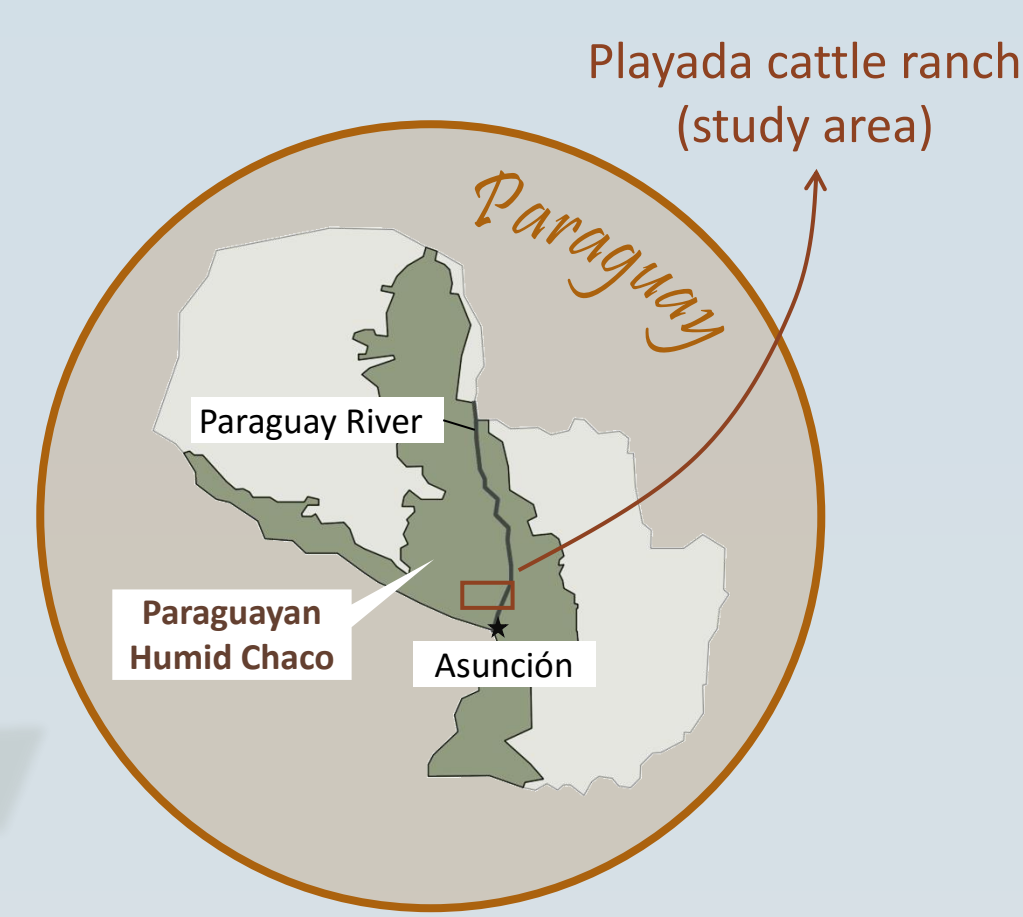
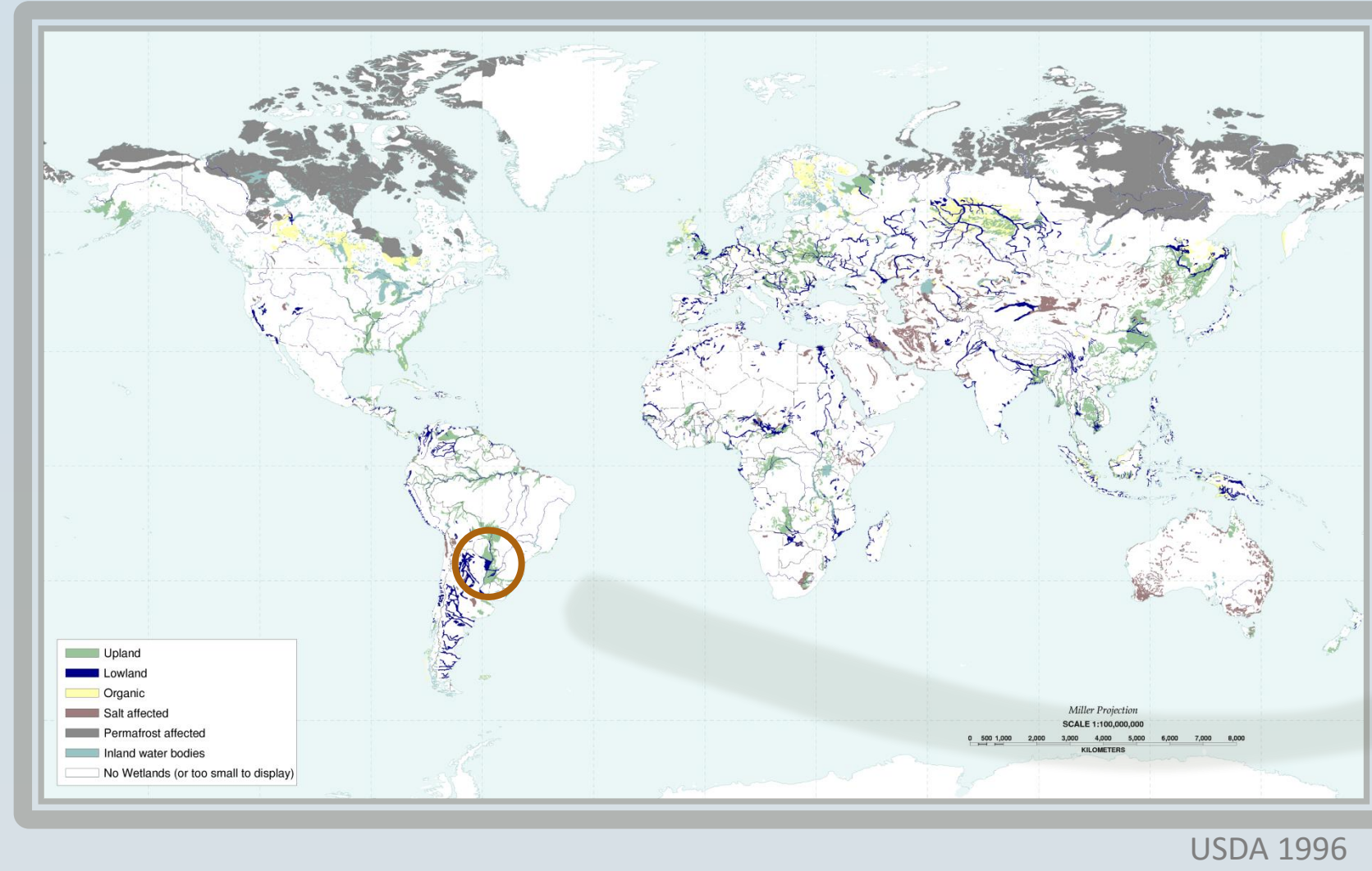


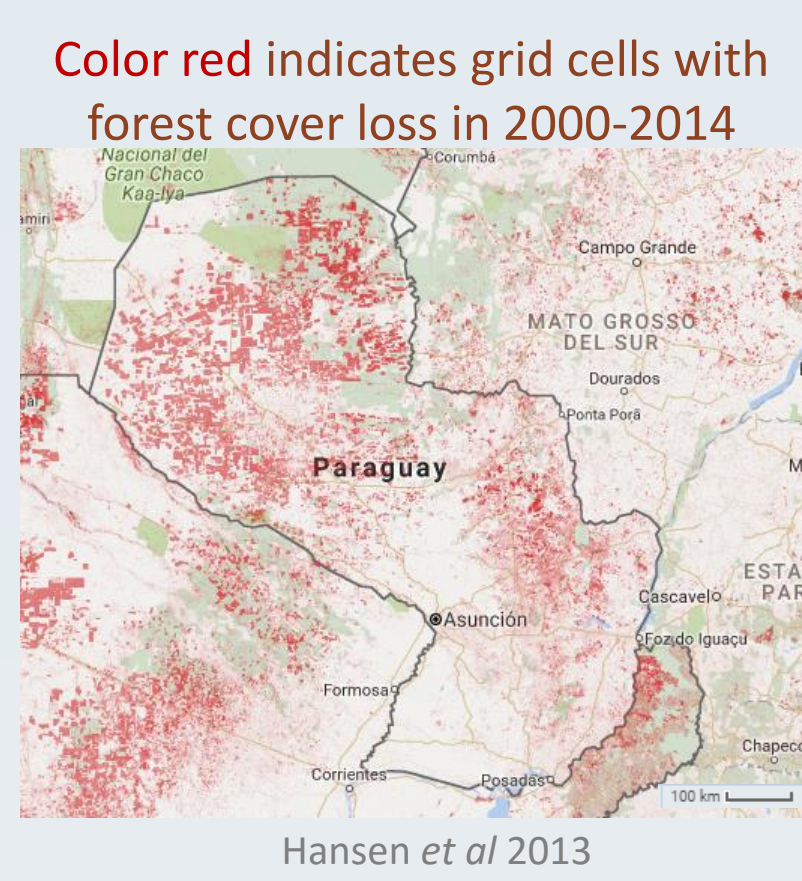
Restoring hydrologic processes in floodable cattle ranches in Paraguay by tailoring revegetation actions

Cruz-Alonso V^{1 2 3}, Musálem K^{1 4}, Viñegla F, Mongil J³, Insfran A^{4 5} & Rey Benayas JM^{2 4}

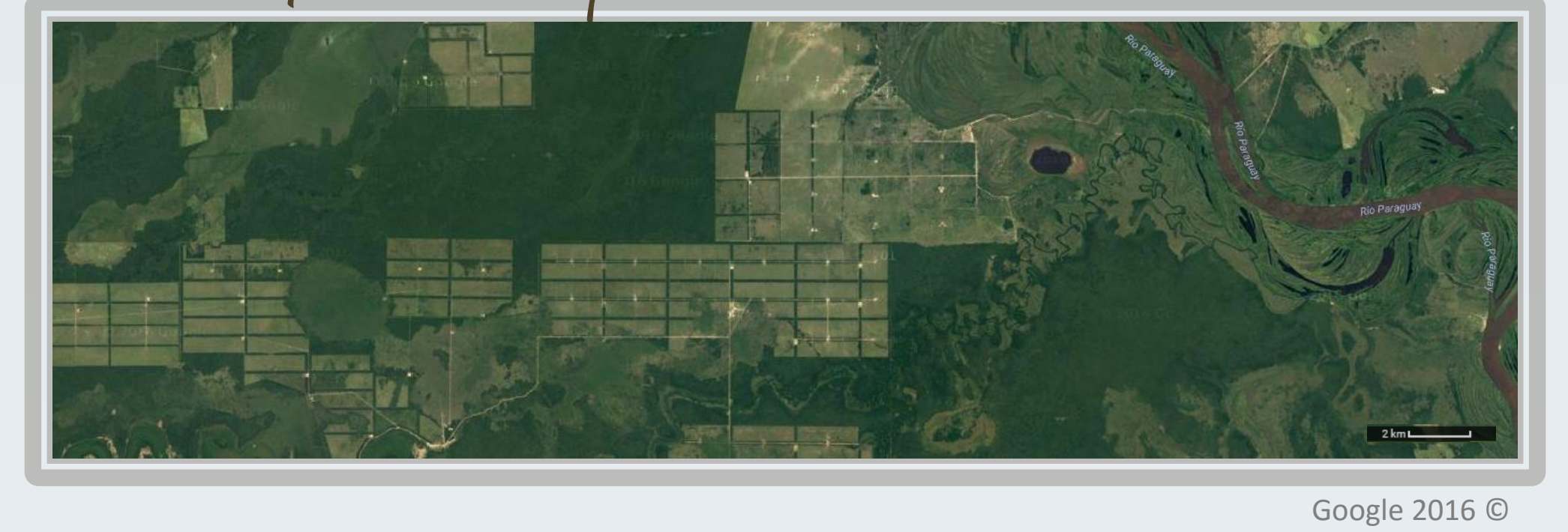
Wetlands around the world



Problems



Deforestation pattern in the Humid Chaco



Reference ecosystem with three vegetation types



Initial hypotheses related to two hydrological processes

Infiltration
Grassland < Forest

Infiltration
Outside the tree < Tree trunk

Evaporation
Grassland > Forest

Infiltration test
One ring ponded
infiltrimeter (Infiltest®)
4 hours test

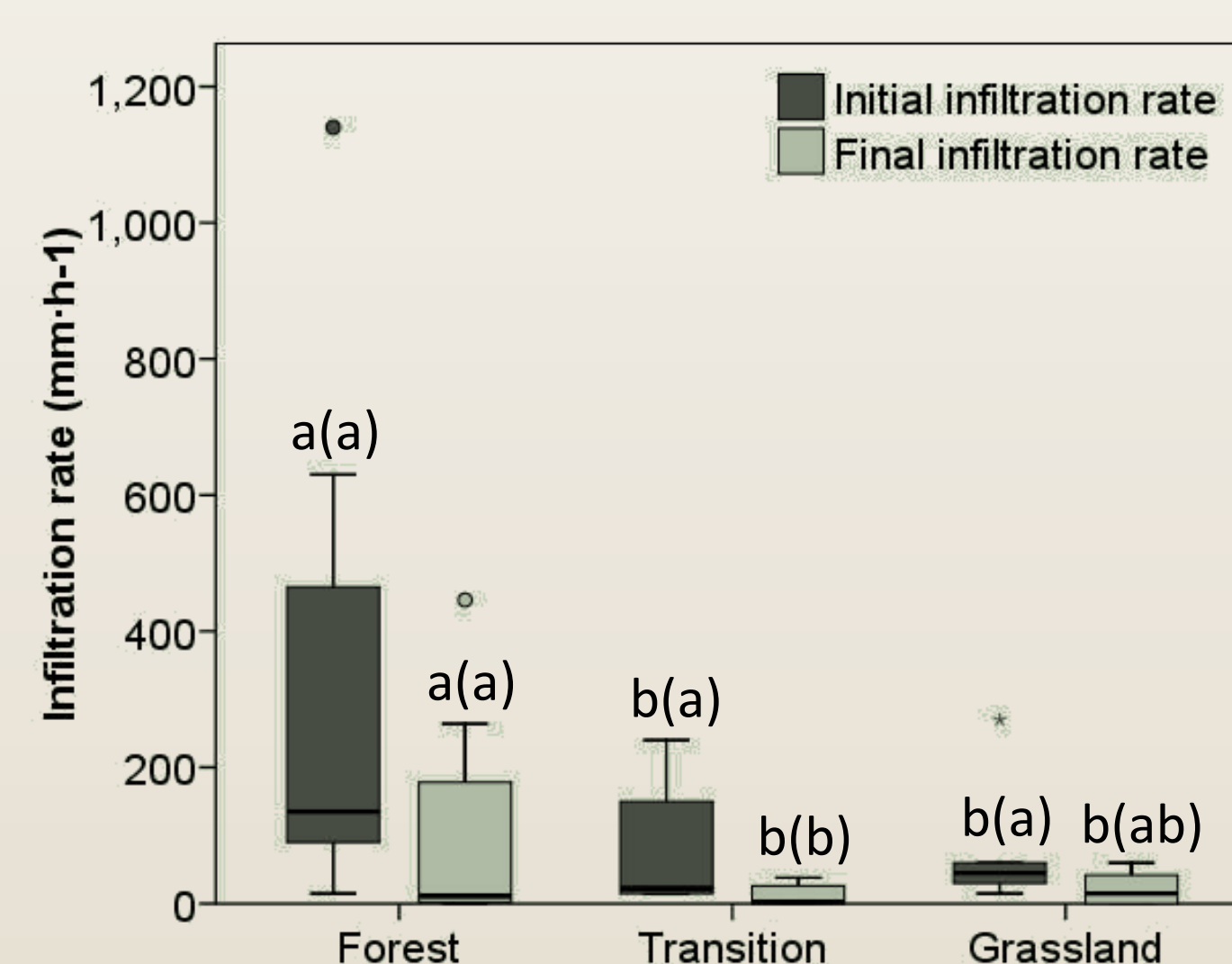


Evaporation test
Floating evaporimeter tank
(Ø= 22 cm; h=8 cm)
5 hours test

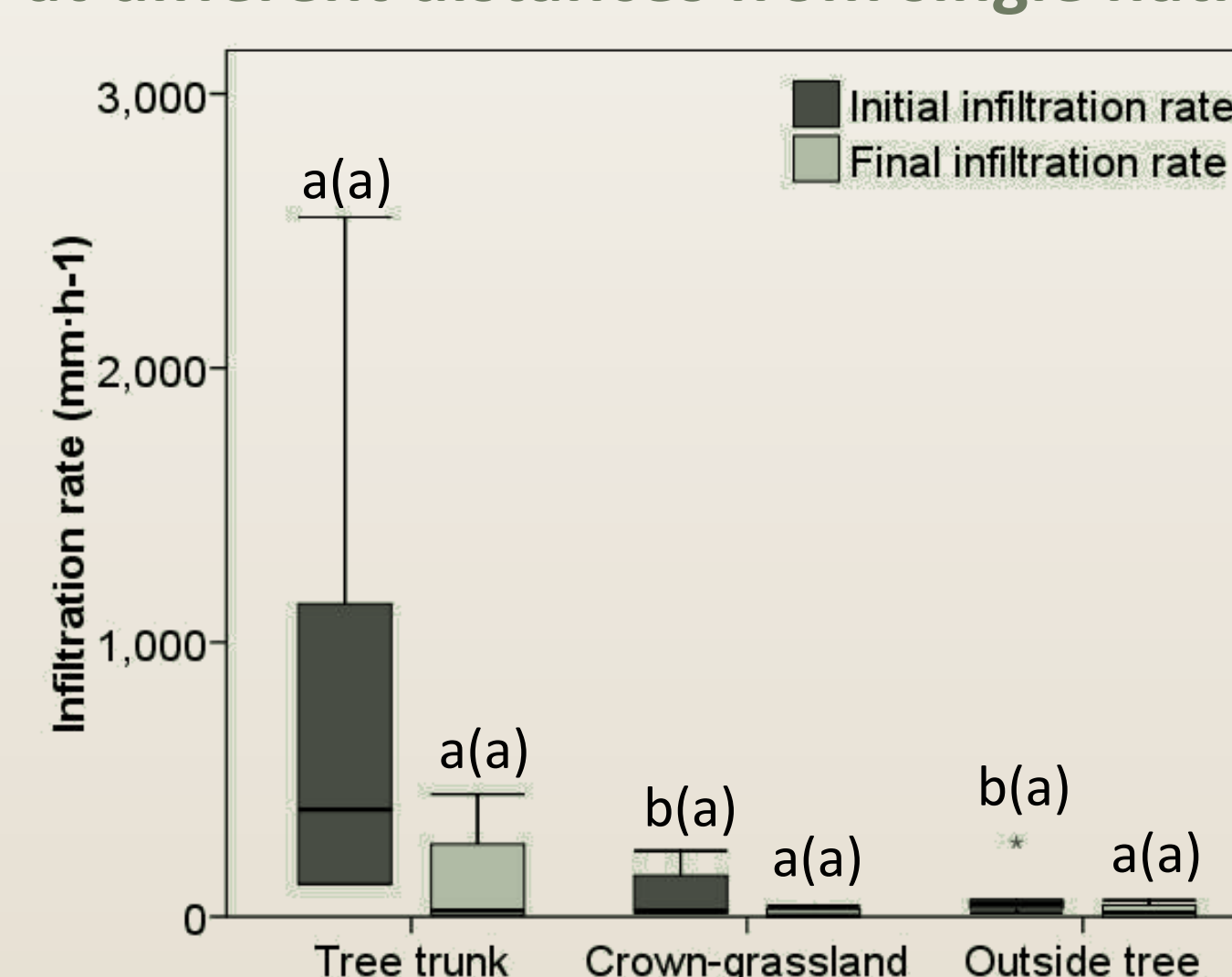


* Means with the same letter are not significantly different with $\alpha = 0.050$ (or $\alpha=0.017$, letter in parenthesis after Bonferroni's correction)

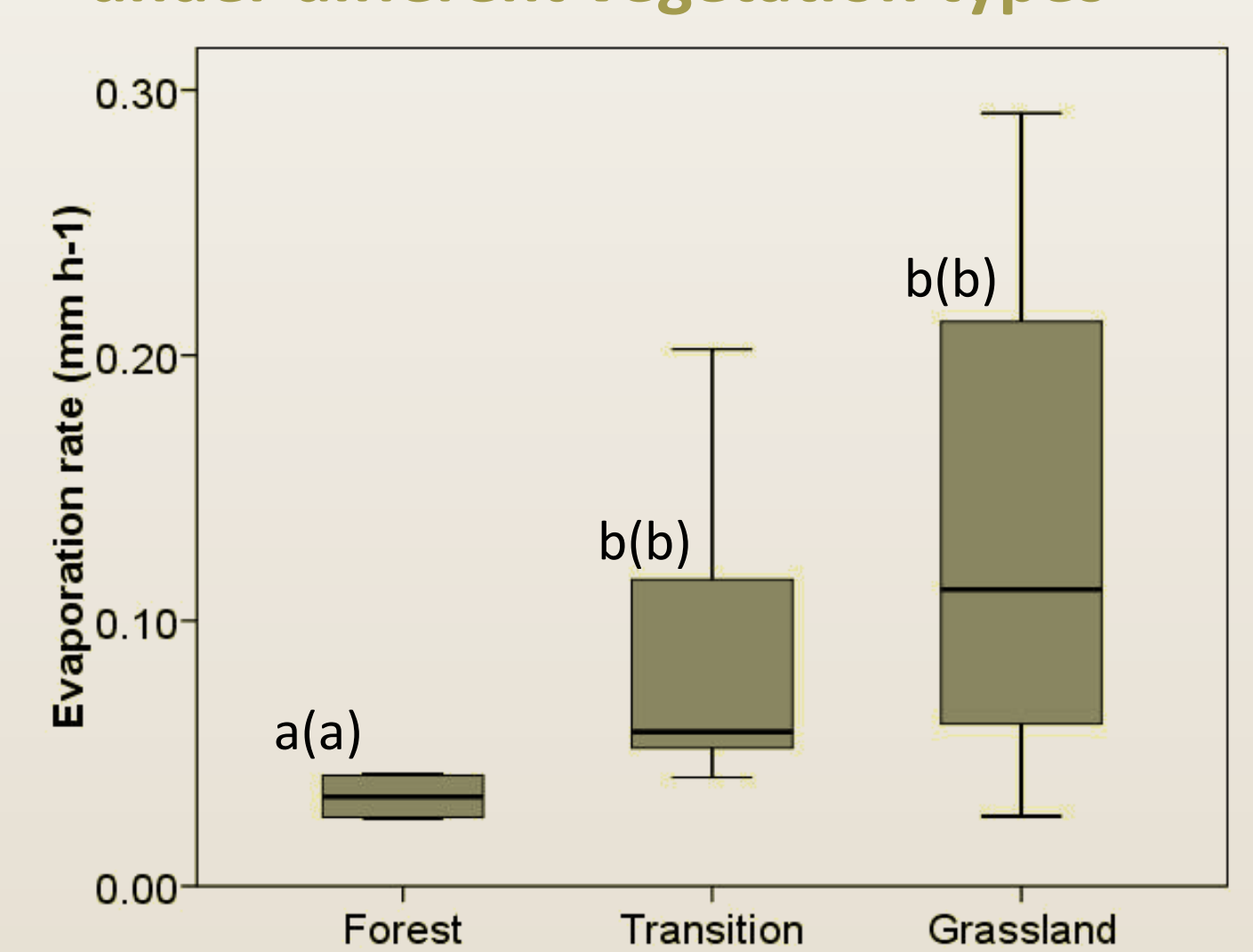
Infiltration under different vegetation types*



Infiltration at different distances from single native trees*



Evaporation under different vegetation types*



Key conclusions

Soil infiltrability and evaporation rates were related to vegetation types in floodable cattle ranches in the Humid Chaco Ecoregion

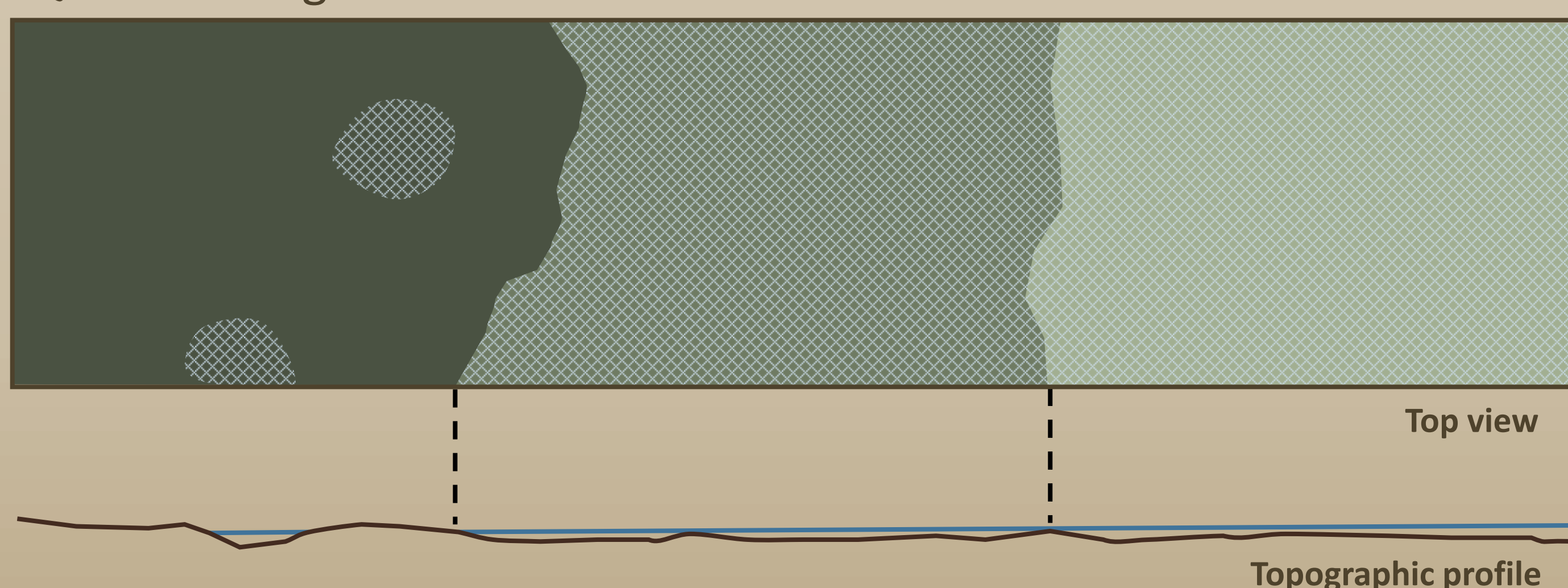
Initial infiltration rates at the tree trunk were marginally greater compared to open areas

The **introduction of scattered forest islets**, or even scattered isolated trees, in deforested pastures could partially enhance water surface processes, reduce flood duration and, as a consequence, **facilitate cattle raising while promoting local biodiversity** linked to native forest patches and trees

Proposed afforestation pattern for improved hydrological services

Forest Pastures Natural grassland Flooded areas Forest islets

a) Initial range



b) Improved range

